CS3000 Algorithms & Data **Practice Quiz 5B**	**Name:** _____ **NUID:** _____

Instructions:

- Please put your phones on silent and in your bag for the duration of the quiz.
- This quiz is designed to last 40 minutes.
- If you have a question, alert a proctor quietly, and we'll be glad to assist.
- If we can't read your answer, we can't grade your answer. Please write legibly.
- Practice version: the format and length match the real quizzes, but these are new problems.

Q1. A drone can carry at most $W = 7$ kg. There are four packages, each either loaded once or not at all: package 1: $(w_1, v_1) = (1, 1)$, package 2: $(w_2, v_2) = (3, 4)$, package 3: $(w_3, v_3) = (4, 5)$, package 4: $(w_4, v_4) = (5, 7)$. Let $K(i, w)$ be the maximum total value using only the first i packages with capacity w.

(a) Write the recurrence for $K(i, w)$, including base cases.

(b) Fill in the table bottom-up.

$i \downarrow \ w \rightarrow$	0	1	2	3	4	5	6	7
$i = 0$								
$i = 1$								
$i = 2$								
$i = 3$								
$i = 4$								

(c) Which packages are loaded in an optimal solution? Show your traceback.

(d) A teammate proposes: "sort packages by value/weight and load greedily." Show that this fails on this instance.

(e) What are the running time and space of the bottom-up algorithm in terms of n and W?

Q1 (continued).

Q2. Let $A =$ `CART` and $B =$ `MARCH`. The allowed operations on A are insertion, deletion, and substitution of a single character, each with cost 1. Let $D(i, j)$ be the edit distance between the first i characters of A and the first j characters of B.

(a) Write the recurrence for $D(i, j)$, including both base cases.

(b) Fill in the table bottom-up.

$A_i \downarrow \ B_j \rightarrow$	0	1 (M)	2 (A)	3 (R)	4 (C)	5 (H)
0						
1 (C)						
2 (A)						
3 (R)						
4 (T)						

(c) What is the edit distance? Trace back from $D(4, 5)$ and list a sequence of operations that achieves it.

Q2 (continued).